Section 1.4 — Inverse Relations — Answer Key

Math 103: College Algebra

1 Writing Inverses

Write the inverse of each relation.

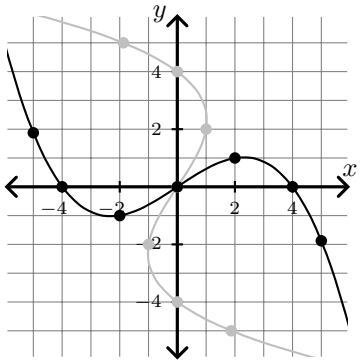
1. $P^{-1} = \{(-6,4),(-2,-10),(-9,-1),(-7,0),(-9,3)\}$
2. $Q^{-1} = \{(5,-7),(3,-2),(9,-2),(-5,10)\}$
3. $R^{-1} = \{(11,-2),(-8,-1),(3,-9),(5,6),(-2,-8)\}$
4. $S^{-1} = \{(-10,-10),(1,5),(-3,3),(9,9),(-6,3)\}$
5. $T^{-1} = \{(x,y) \mid x = 3y-4\}$
6. $U^{-1} = \{(x,y) \mid y^2 = x^3\}$
7. $V^{-1} = \{(x,y) \mid y = x^5 + y^2x + x^3\}$
8. $W^{-1} = \{(x,y) \mid y^2 + x^2 = 25\}$

2 Graphing Inverses

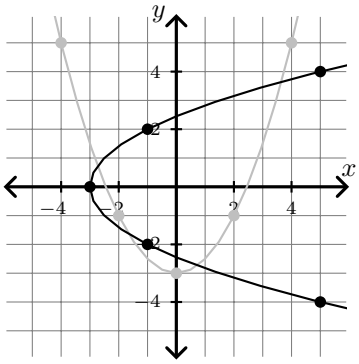
Several relations are graphed below. Sketch the graphs of their inverses. *Hint: Used the points marked to guide your sketch.*

(Note: the original graph is shown grayed out for comparison.)

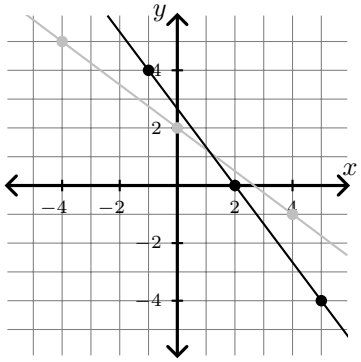
9.



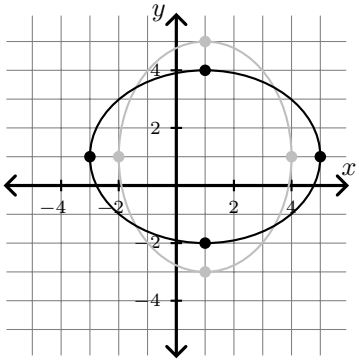
10.



11.



12.



3 Domain and Range of the Inverse

13. $\operatorname{Dom} R = \{-2,3\}$, $\operatorname{Rng} R = \{4,5,6\}$, $\operatorname{Dom} R^{-1} = \{4,5,6\}$, $\operatorname{Rng} R^{-1} = \{-2,3\}$
14. $\operatorname{Dom} S^{-1} = \operatorname{Rng} S = (-\infty, 11]$, $\operatorname{Rng} S^{-1} = \operatorname{Dom} S = [-5, 2)$
15. $\operatorname{Rng} T = \operatorname{Dom} T^{-1} = \{x \mid x \neq -1\}$, $\operatorname{Rng} T^{-1} = \operatorname{Dom} T = \{x \mid x \neq 5\}$

4 Identifying One-to-One

For each relation below, decide which of the following applies to the relation. (This is multiple choice.)

a. A function (but not one-to-one)
b. One-to-one but not a function
c. A one-to-one function
d. Not one-to one or a function

16. a - a function (but not one-to-one)
17. d - not one-to-one or a function
18. a - a function (but not one-to-one)
19. b - one-to-one but not a function
20. d - not one-to-one or a function
21. a - a function (but not one-to-one)
22. c - a one-to-one function
23. b - one-to-one but not a function

For each relation below, determine which of the four ordered pairs given are in the relation. Use this to decide what kind of relation this is. (This is multiple choice.)

a. A function (but not one-to-one)
b. One-to-one but not a function
c. A one-to-one function
d. Not one-to one or a function

The pairs are chosen so that, if the relation is *not* one-to-one, the given ordered pairs will demonstrate this.

24. c - a one-to-one function
25. d - not one-to-one or a function
26. a - a function (but not one-to-one)
27. b - one-to-one but not a function

5 Inverse Functions

The function $h = \{(-4,-7),(2,2),(4,8),(8,-9)\}$ is one-to-one. Find each of the following, or write "undefined."

28. $h^{-1}(-7) = -4$
29. $h^{-1}(8)$ is undefined
30. $h^{-1}(4)$ is undefined
31. $h^{-1}(2) = 2$

For each function given below, write the inverse function. Answer in function notation.

32. $f^{-1}(x) = \frac{1}{2}x + 4$
33. $g^{-1}(x) = \frac{1}{5}x - 3$

The function $k(x) = \dfrac{x-3}{2x+1}$ is one-to-one. Evaluate each of the following.

34. $k(-4) = 1$
35. $k(-1) = 4$
36. $k(0) = -3$
37. $k(3) = 0$

Based on the values of $k(x)$ you found, evaluate each of the following.

38. $k^{-1}(-3) = 0$
39. $k^{-1}(0) = 3$
40. $k^{-1}(1) = -4$
41. $k^{-1}(4) = -1$